

Week 3: Seller and Inventory Management system

Scenario

The railway reservation system manages trains operated across multiple routes and stations. The system should maintain **train details**, track the **routes and stations associated with each train**, and monitor **seat availability** to ensure passengers can book available seats for their journey.

1. Create Seller and Inventory Tables

```
SQL> CREATE TABLE Seller(Seller_ID VARCHAR(7), Seller_Name VARCHAR(25), Email VARCHAR(30), Phone number(10), Address VARCHAR(45));  
Table created.
```

```
SQL> CREATE TABLE Inventoryy(Inventory_ID VARCHAR(7),  
2 Product_ID VARCHAR(7), Seller_ID VARCHAR(7),  
3 Stock_Quantity number, Stock_Status VARCHAR(11),  
4 Last_Updated DATE);
```

```
Table created.
```

2. Establish Relationships

Seller and Product One seller can supply multiple products.

One product can be supplied by one or multiple sellers.

Relationship:

Seller → Product is one to many

Product and Inventory Each product has inventory information.

Inventory tracks available stock.

Relationship:

Product → Inventory is one to one

```
SQL> ALTER TABLE Seller ADD CONSTRAINT PK_Seller PRIMARY KEY(Seller_ID);  
Table altered.  
  
SQL> ALTER TABLE Inventory ADD CONSTRAINT PK_Inv PRIMARY KEY(Inventory_ID);  
Table altered.
```

Foreign Key for Inventory Table:

```
SQL> ALTER TABLE Inventory ADD  
2 CONSTRAINT FK_Prod FOREIGN KEY(Product_ID) REFERENCES Product(Product_I  
D);  
Table altered.  
  
SQL> ALTER TABLE Inventory ADD  
2 CONSTRAINT FK_Seller FOREIGN KEY(Seller_ID) REFERENCES Seller(Seller_I  
D);  
Table altered.
```

Constraints to make Unique Email ID:

```
SQL> ALTER TABLE Seller ADD  
2 CONSTRAINT Un_mail UNIQUE(Email);  
Table altered.
```

To make Email NOT NULL

```
SQL> ALTER TABLE Seller MODIFY Email NOT NULL;  
Table altered.
```

Add Constraint to Check the Stock Quantity is Positive

```
SQL> ALTER TABLE Inventory ADD  
2 CONSTRAINT CK_Quan CHECK(Stock_Quantity >=0);  
Table altered.
```

Insert Values to Seller Table

```
SQL> INSERT INTO Seller VALUES('&SellerID','&SellerName','&email','&phone','&Address');
Enter value for sellerid: S00001
Enter value for sellername: Vani Enterprises
Enter value for email: mdvanienter@gmail.com
Enter value for phone: 8768967547
Enter value for address: 1, Anna Nagar, chennai
old 1: INSERT INTO Seller VALUES('&SellerID','&SellerName','&email','&phone','&Address')
new 1: INSERT INTO Seller VALUES('S00001','Vani Enterprises','mdvanienter@gmail.com','8768967547','1, Anna Nagar, chennai')

1 row created.
```

```
SQL> /
Enter value for sellerid: S00002
Enter value for sellername: Sir Traders
Enter value for email: sritraders@gmail.com
Enter value for phone: 9856745689
Enter value for address: T Nagar ,chennai
old 1: INSERT INTO Seller VALUES('&SellerID','&SellerName','&email','&phone','&Address')
new 1: INSERT INTO Seller VALUES('S00002','Sir Traders','sritraders@gmail.com','9856745689','T Nagar ,chennai')

1 row created.
```

```
SQL> /
Enter value for sellerid: S0003
Enter value for sellername: Kumar Stores
Enter value for email: kumarstores@gmail.com
Enter value for phone: 7865098432
Enter value for address: Adyar, Chennai
old 1: INSERT INTO Seller VALUES('&SellerID','&SellerName','&email','&phone','&Address')
new 1: INSERT INTO Seller VALUES('S0003','Kumar Stores','kumarstores@gmail.com','7865098432','Adyar, Chennai')

1 row created.
```

Activate Windows
Go to Settings to activate Windows.

```
SQL> SELECT * FROM Seller;
```

SELLER_	SELLER_NAME	EMAIL	PHONE
000001	Vani Enterprises Anna Nagar, chennai	mdvanienter@gmail.com	8768967547
000002	Sir Traders Nagar ,chennai	sritraders@gmail.com	9856745689
000003	Kumar Stores Adyar, Chennai	kumarstores@gmail.com	7865098432
000004	ABC Suppliers Velachery, Chennai	abcsuppliers@gmail.com	9876543213
000005	Green Mart Tambaram, Chennai	greenmart@gmail.com	9865709234
000006	Ravi Agencies Guindy, Chennai	raviagencies@gmail.com	8945672345

```
6 rows selected.
```

Activate Windows

Go to Settings to activate Windows.

Insert Values to Inventory Table

```
SQL> INSERT INTO Inventory VALUES('&InvID','&ProdID','&Sell',&Stock,'&Stock_
status',TO_DATE('&Last_upd','YYYY-MM-DD'));
Enter value for invid: Inv003
Enter value for prodid: P003
Enter value for sell: S000003
Enter value for stock: 6
Enter value for stock_status: Unavailable
Enter value for last_upd: 2026-09-15
old 1: INSERT INTO Inventory VALUES('&InvID','&ProdID','&Sell',&Stock,'&St
ock_status',TO_DATE('&Last_upd','YYYY-MM-DD'))
new 1: INSERT INTO Inventory VALUES('Inv003','P003','S000003',6,'Unavailab
le',TO_DATE('2026-09-15','YYYY-MM-DD'))

1 row created.
```

```
SQL> INSERT INTO Seller
  2 VALUES ('S000004', 'Seller 4', 'seller4@gmail.com', 9876543214, 'Madura
i');

1 row created.
```

```
SQL> INSERT INTO Inventory VALUES('&InvID','&ProdID','&Sell',&Stock,'&Stock_
status',TO_DATE('&Last_upd','YYYY-MM-DD'));
Enter value for invid: Inv004
Enter value for prodid: P004
Enter value for sell: S000004
Enter value for stock: 8
Enter value for stock_status: Available
Enter value for last_upd: 2026-09-16
old 1: INSERT INTO Inventory VALUES('&InvID','&ProdID','&Sell',&Stock,'&St
ock_status',TO_DATE('&Last_upd','YYYY-MM-DD'))
new 1: INSERT INTO Inventory VALUES('Inv004','P004','S000004',8,'Available
',TO_DATE(' 2026-09-16','YYYY-MM-DD'))

1 row created.
```

```
SQL>
```

```
SQL> SELECT * FROM Inventory;
```

INVENTO	PRODUC	SELLER_	STOCK_QUANTITY	STOCK_STATU	LAST_UPDA
Inv003	P003	S000003	6	Unavailable	15-SEP-26
Inv004	P004	S000004	8	Available	16-SEP-26
Inv005	P005	S000005	4	Available	17-SEP-26

```
SQL> SELECT * FROM Seller;
```

Report 1: Seller-wise Product Report Display:

- Seller name

- Number of products supplied
- Available stock

```
SQL> SELECT s.Seller_Name,
2         COUNT(i.Product_ID) AS Number_of_Products,
3         SUM(i.Stock_Quantity) AS Available_Stock
4 FROM Seller s
5 JOIN Inventory i ON s.Seller_ID = i.Seller_ID
6 WHERE i.Stock_Status = 'Available'
7 GROUP BY s.Seller_Name;
```

SELLER_NAME	NUMBER_OF_PRODUCTS	AVAILABLE_STOCK
Seller 5	1	4
Seller 4	1	8

Use Joins to get details from Seller and Inventory Table.

Report 2: Stock Availability Report Display:

- Product name
- Stock quantity
- Stock status

```
SQL> SELECT p.Product_Name,
2         i.Stock_Quantity,
3         i.Stock_Status
4 FROM Product p
5 JOIN Inventory i ON p.Product_ID = i.Product_ID;
```

PRODUCT_NAME	STOCK_QUANTITY	STOCK_STATU
Redmi Case	6	Unavailable
OnePlus Case	8	Available
Vivo Case	4	Available

Report 3: Inventory Analysis Find:

- Total products available

```
SQL> SELECT COUNT(*) AS Total_Available_Products
  2   FROM Inventory
  3   WHERE Stock_Status = 'Available';
```

```
TOTAL_AVAILABLE_PRODUCTS
-----
                        2
```

- Products out of stock

```
SQL> SELECT COUNT(*) AS OutOfStock_Products
  2   FROM Inventory
  3   WHERE Stock_Status = 'Out of Stock';
```

```
OUTOFSTOCK_PRODUCTS
-----
                        0
```

- Highest stocked products

```
SQL> SELECT p.Product_Name, i.Stock_Quantity
  2   FROM Product p
  3   JOIN Inventory i ON p.Product_ID = i.Product_ID
  4   WHERE i.Stock_Quantity = (SELECT MAX(Stock_Quantity) FROM Inventory);
```

```
PRODUCT_NAME
-----
-----
STOCK_QUANTITY
-----
OnePlus Case
                8
```

- Average inventory quantity

```
SQL> SELECT AVG(Stock_Quantity) AS Average_Inventory
  2   FROM Inventory;
```

```
AVERAGE_INVENTORY
-----
                        6
```